

Lab 3: Architectural Pattern Justification

PES1UG24CS585 | Domain & SSL Certificate Expiry Alert System | Developer Tools & IT Operations

Architecture Selection: "We chose Layered Architecture for the Domain & SSL Certificate Expiry Alert System."

Reason 1: Clear Separation of Concerns

The system has three naturally distinct concerns: presenting expiry status to the SysAdmin and Security Officer (Presentation Layer), the certificate-scanning and alerting logic (Business Layer), and the persisted record of monitored domains, certificates, and alert history (Data Layer). A Layered Architecture keeps these concerns isolated, so the scanning logic can be modified (e.g., adding a new certificate authority check) without touching the dashboard or the storage layer.

Reason 2: Matches the System's Scale and Workflow

The alert system serves a predictable, linear workflow: a scheduled trigger initiates a scan, the scan checks domain/SSL expiry, results are stored, and notifications are sent if a certificate is nearing expiry. This maps cleanly onto layers without the operational overhead of independently deployed microservices, which would be unnecessary for an internal tool monitoring a bounded set of organizational domains.

Security Advantage

Because the Presentation Layer (Admin Dashboard) never accesses the Domain & Certificate Database directly, all requests must pass through the Business Layer's Certificate Scanner component, which can enforce authentication, role checks (SysAdmin vs. Security Officer), and input validation before any sensitive certificate data is exposed. This confines the database attack surface to a single, auditable layer rather than distributing data access across the UI.

Performance Benefit

Since certificate checks are periodic rather than continuous or high-frequency, the layered design lets the Scheduler, Certificate Scanner, and Alert Notification components communicate through fast in-process calls rather than network calls between independently deployed services. This keeps the delay between detecting a near-expiry certificate and delivering an alert to the SysAdmin or Security Officer to a minimum, without the network latency overhead a microservices approach would introduce for this scale of system.